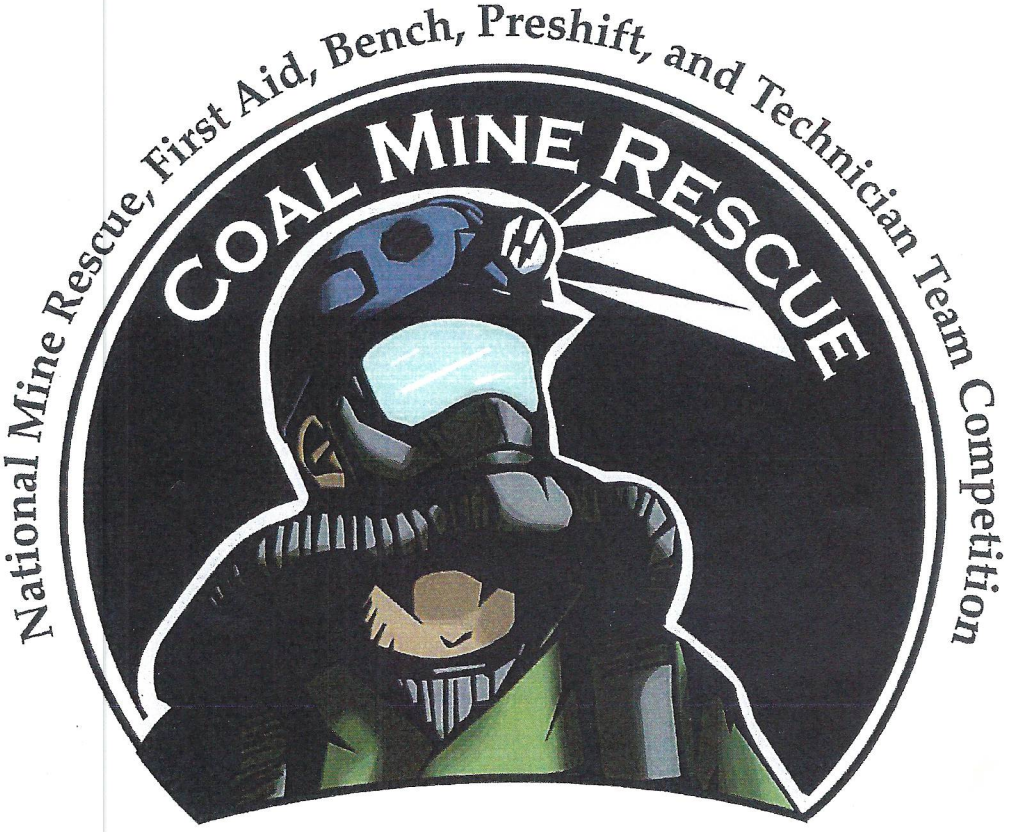
2026 National Mine Rescue Contest

Coal Day 1

Mine Rescue

Judges Packet



2026 National Mine Rescue Contest

Thank you for answering our call for help here at the Mystery Mine. You are located at the fresh air base on the Wears Valley Submains. We are currently driving a three-entry panel for a future ventilation project. Our maps were last updated two days ago, and we think we are getting close to punching through into our Ogle Fork Mains.

Early this morning, an eight-man crew entered this section to do some routine maintenance and haul a few loads of coal. At approximately 3:30 AM, the scoop operator on this crew returned to the surface. He described what he saw as a "blazing fury" and said that he couldn't get to the section for smoke. Attempts to contact the crew have been unsuccessful.

Our mine rescue team has explored up to this point and has asked to be relieved. We are asking you to continue exploration and help account for our missing miners. State and Federal Agencies have been notified and are on site. Backup mine rescue teams are available if needed.

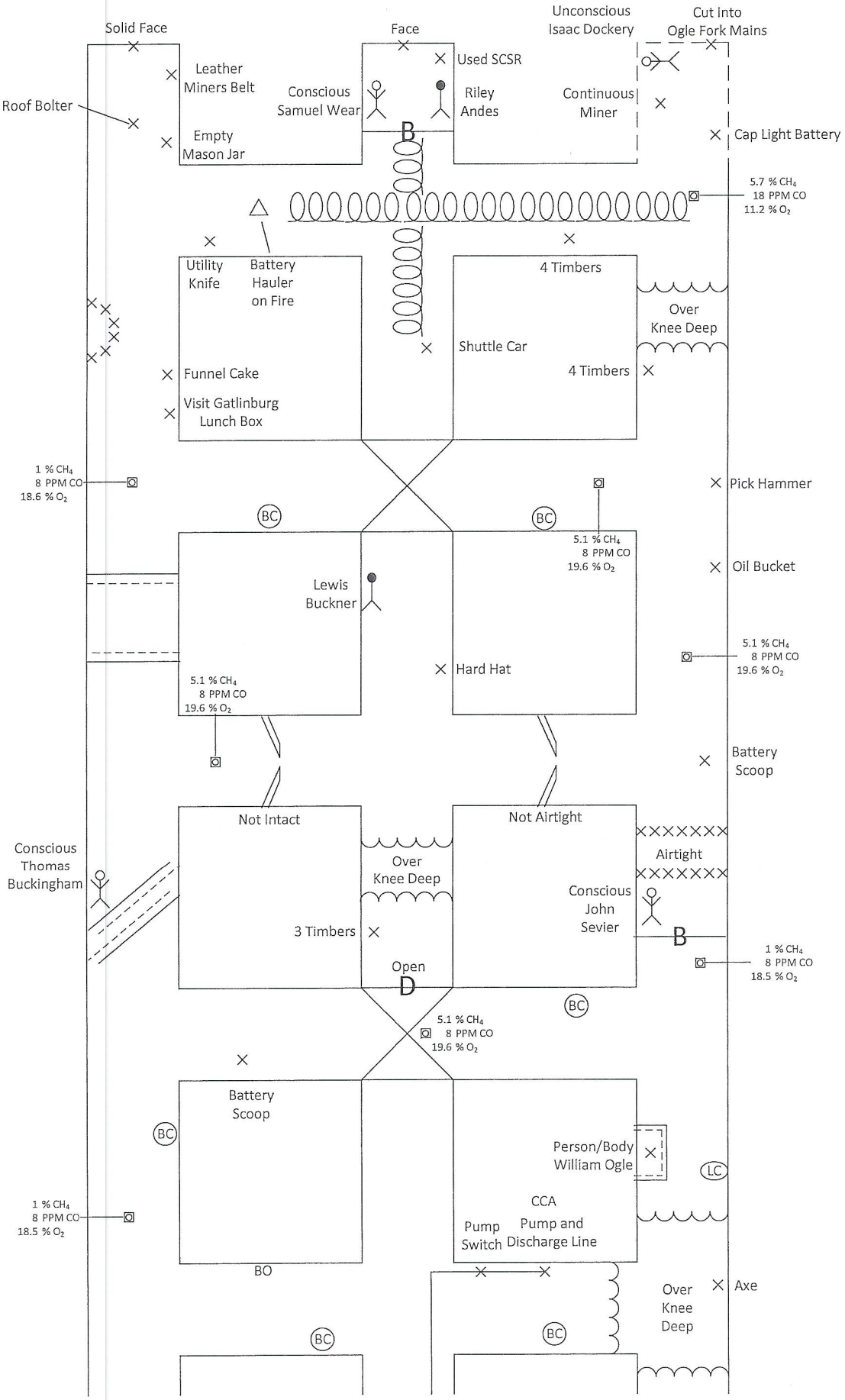
All underground power to the mine has been de-energized and locked out. This mine is ventilated with a blowing fan on the surface that is currently off. Requests to energize electrical circuits and start the fan must be made through the Superintendent. Once the fan is started, it cannot be stopped, stalled, or reversed. Typical ventilation for this section has intake air coming up the #2 entry and the #1 and #3 entries are the returns. We have a history of methane, bad roof, and water at this mine.

You have 75 minutes to complete the problem. Thanks for coming to help and good luck!

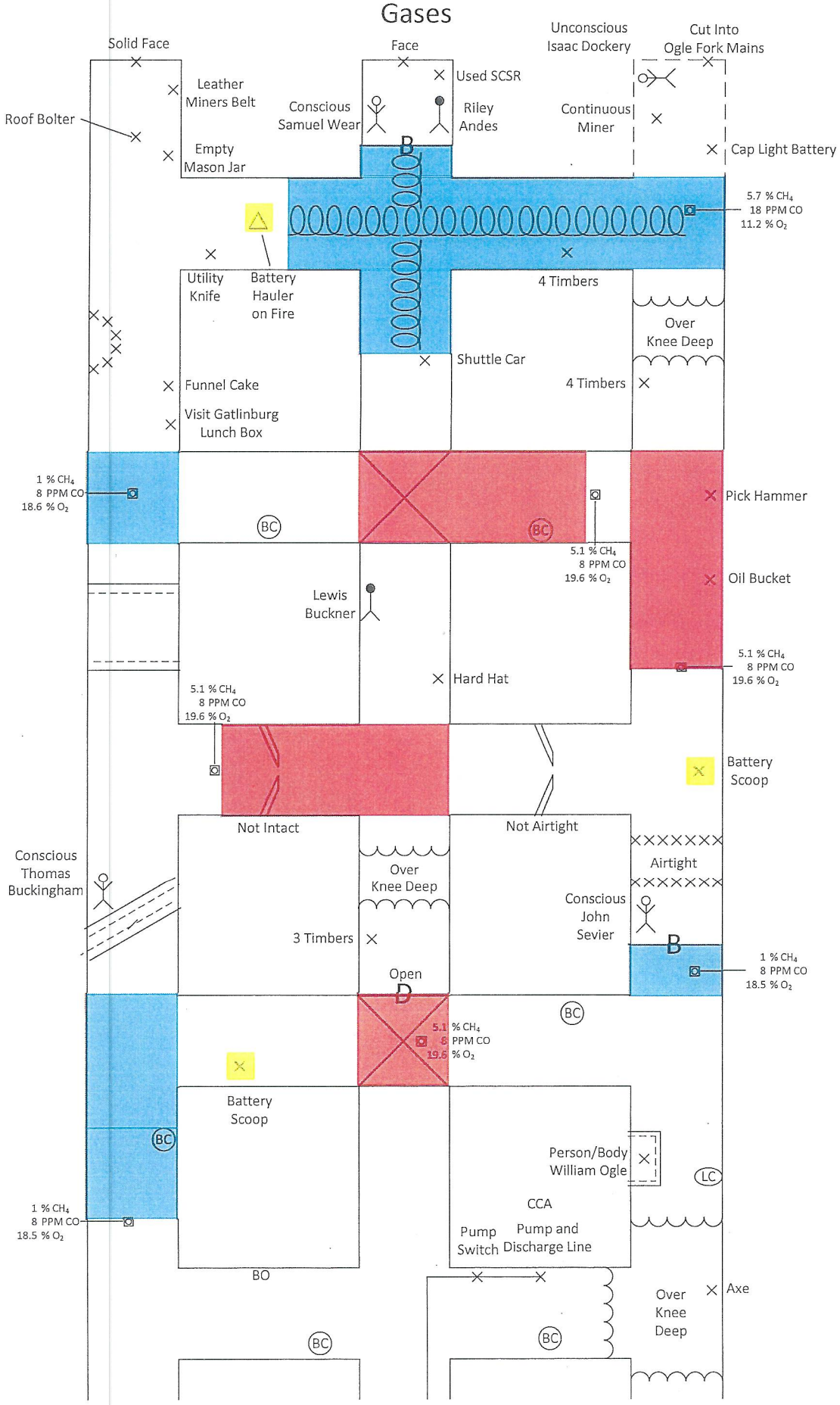
Written Instructions

- Account for all missing miners
- Bring all survivors to the fresh air base
- Explore all areas of the mine that can be explored safely
- Team can only carry two brattice cloths at a time
- The name and location of any bodies discovered shall be reported to the Superintendent prior to the team leaving that team stop
- Team cannot leave the field with any materials that they picked up during the problem

Problem


Solid Face
Face
Unconscious
Isaac Dockery
Cut Into
Ogle Fork Mains
Used SCSR
Leather
Miners Belt
Conscious
Samuel Wear
Riley
Andes
Continuous
Miner
Roof Bolter
Empty
Mason Jar
B
Cap Light Battery
5.7 % CH4
18 PPM CO
11.2 % O2
Utility
Knife
Battery
Hauler
on Fire
4 Timbers
Over
Knee Deep
Shuttle Car
Funnel Cake
4 Timbers
Visit Gatlinburg
Lunch Box
1 % CH4
8 PPM CO
18.6 % O2
Pick Hammer
BC
BC
5.1 % CH4
8 PPM CO
19.6 % O2
Oil Bucket
Lewis
Buckner
5.1 % CH4
8 PPM CO
19.6 % O2
Hard Hat
5.1 % CH4
8 PPM CO
19.6 % O2
Battery
Scoop
Not Intact
Not Airtight
Conscious
Thomas
Buckingham
Over
Knee Deep
Airtight
Conscious
John
Sevier
3 Timbers
B
Open
D
1 % CH4
8 PPM CO
18.5 % O2
BC
5.1 % CH4
8 PPM CO
19.6 % O2
Battery
Scoop
BC
Person/Body
William Ogle
LC
CCA
Pump
Switch
Pump and
Discharge Line
1 % CH4
8 PPM CO
18.5 % O2
BO
Over
Knee
Deep
Axe
BC
BC

Extent of Gases



Patient #1

Thomas Buckingham

"Help, get me out, there is bad roof in front of me."

Patient #2

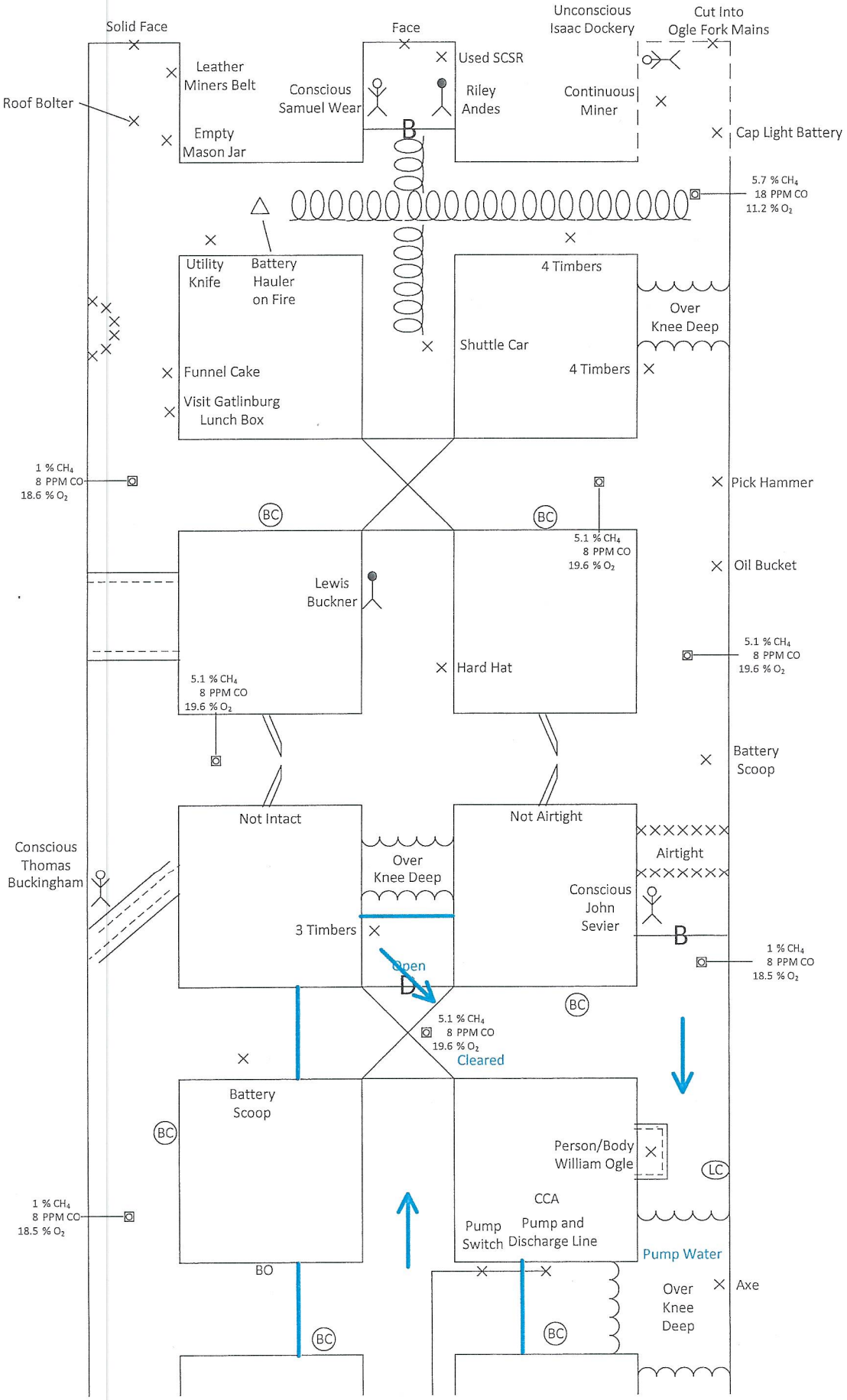
John Sevier

"Help, get me out, it is airtight behind me."

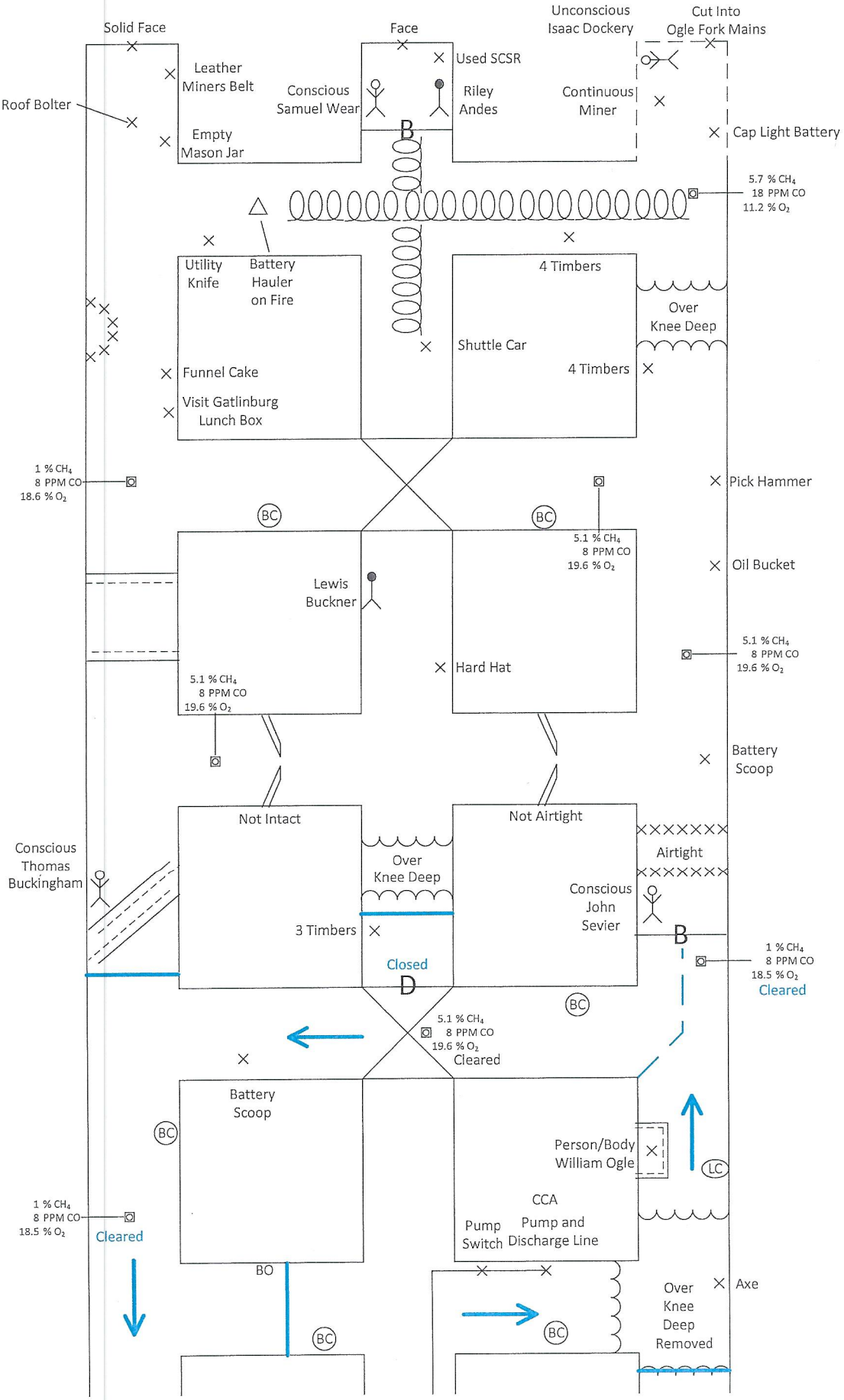
Patient #3
Samuel Wear

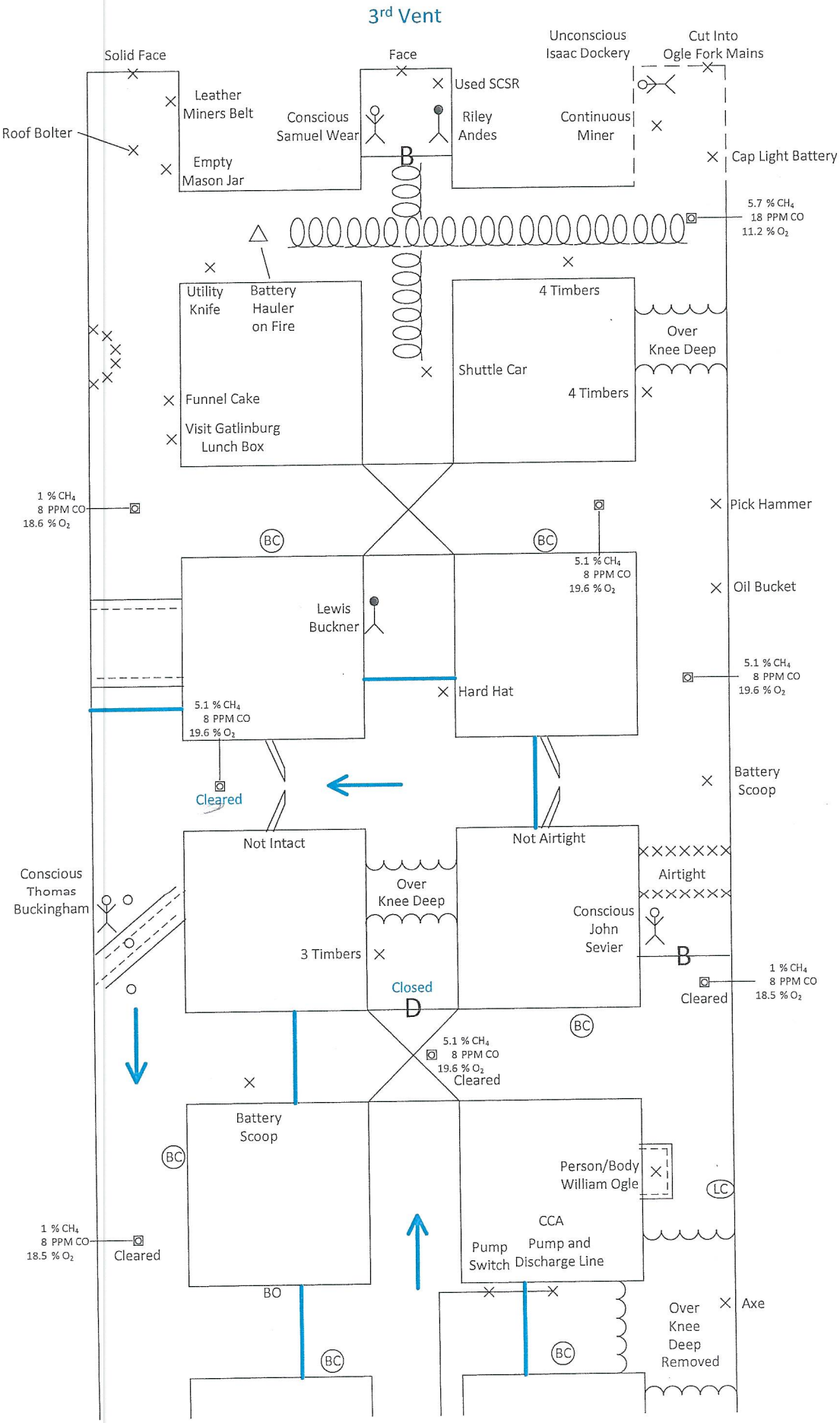
"Help, get me out."

1ST Vent

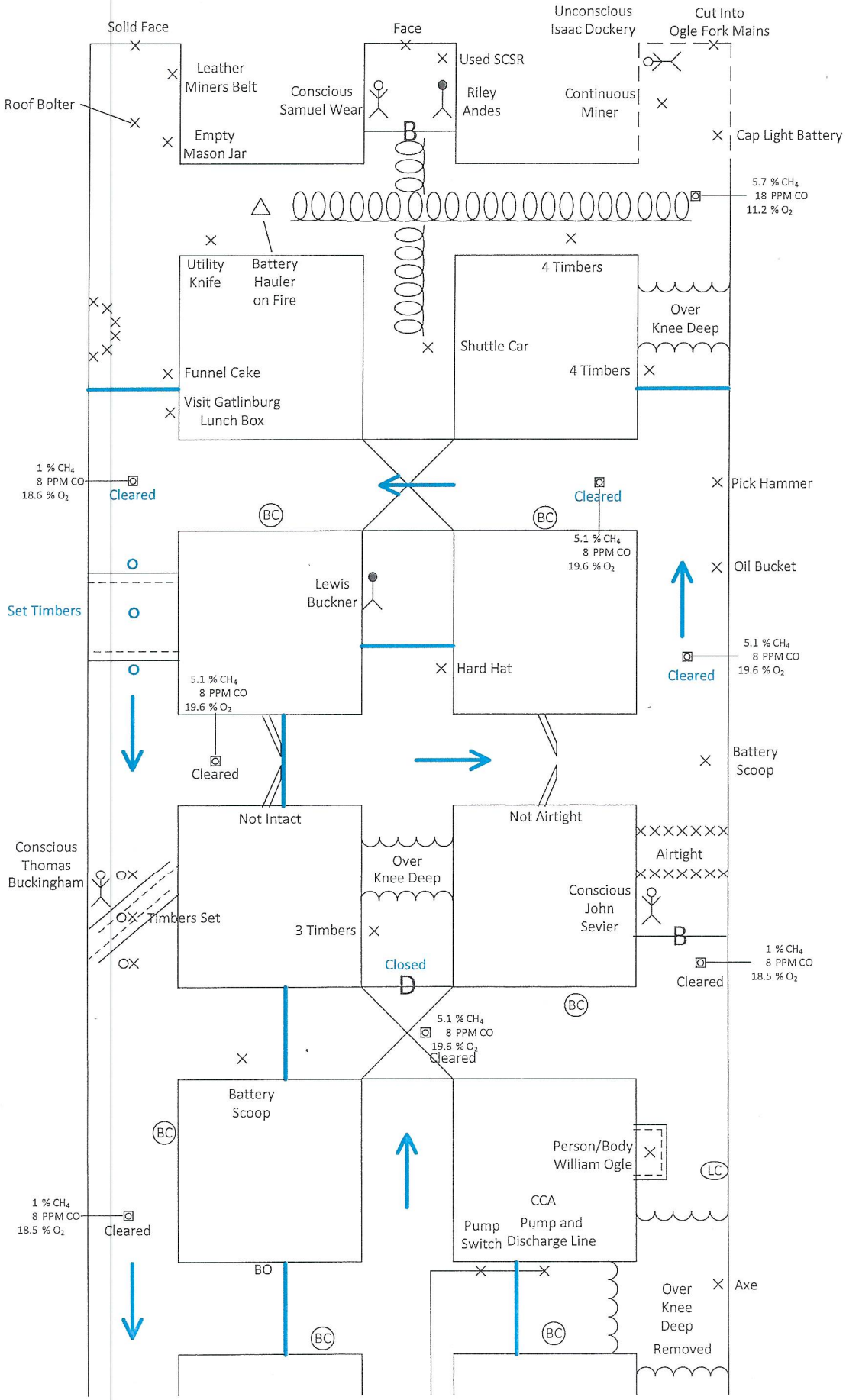

Solid Face
Face
Unconscious
Isaac Dockery
Cut Into
Ogle Fork Mains
Leather
Miners Belt
Used SCSR
Roof Bolter
Conscious
Samuel Wear
Riley
Andes
Continuous
Miner
Empty
Mason Jar
B
Cap Light Battery
5.7 % CH4
18 PPM CO
11.2 % O2
Utility
Knife
Battery
Hauler
on Fire
4 Timbers
Over
Knee Deep
Shuttle Car
Funnel Cake
4 Timbers
Visit Gatlinburg
Lunch Box
1 % CH4
8 PPM CO
18.6 % O2
BC
BC
Pick Hammer
5.1 % CH4
8 PPM CO
19.6 % O2
Oil Bucket
Lewis
Buckner
5.1 % CH4
8 PPM CO
19.6 % O2
Hard Hat
5.1 % CH4
8 PPM CO
19.6 % O2
Battery
Scoop
Not Intact
Not Airtight
Conscious
Thomas
Buckingham
Over
Knee Deep
Airtight
Conscious
John
Sevier
3 Timbers
B
1 % CH4
8 PPM CO
18.5 % O2
Open
D
BC
5.1 % CH4
8 PPM CO
19.6 % O2
Cleared
Battery
Scoop
BC
Person/Body
William Ogle
LC
CCA
1 % CH4
8 PPM CO
18.5 % O2
Pump
Switch
Pump and
Discharge Line
Pump Water
BO
Over
Knee
Deep
Axe
BC
BC

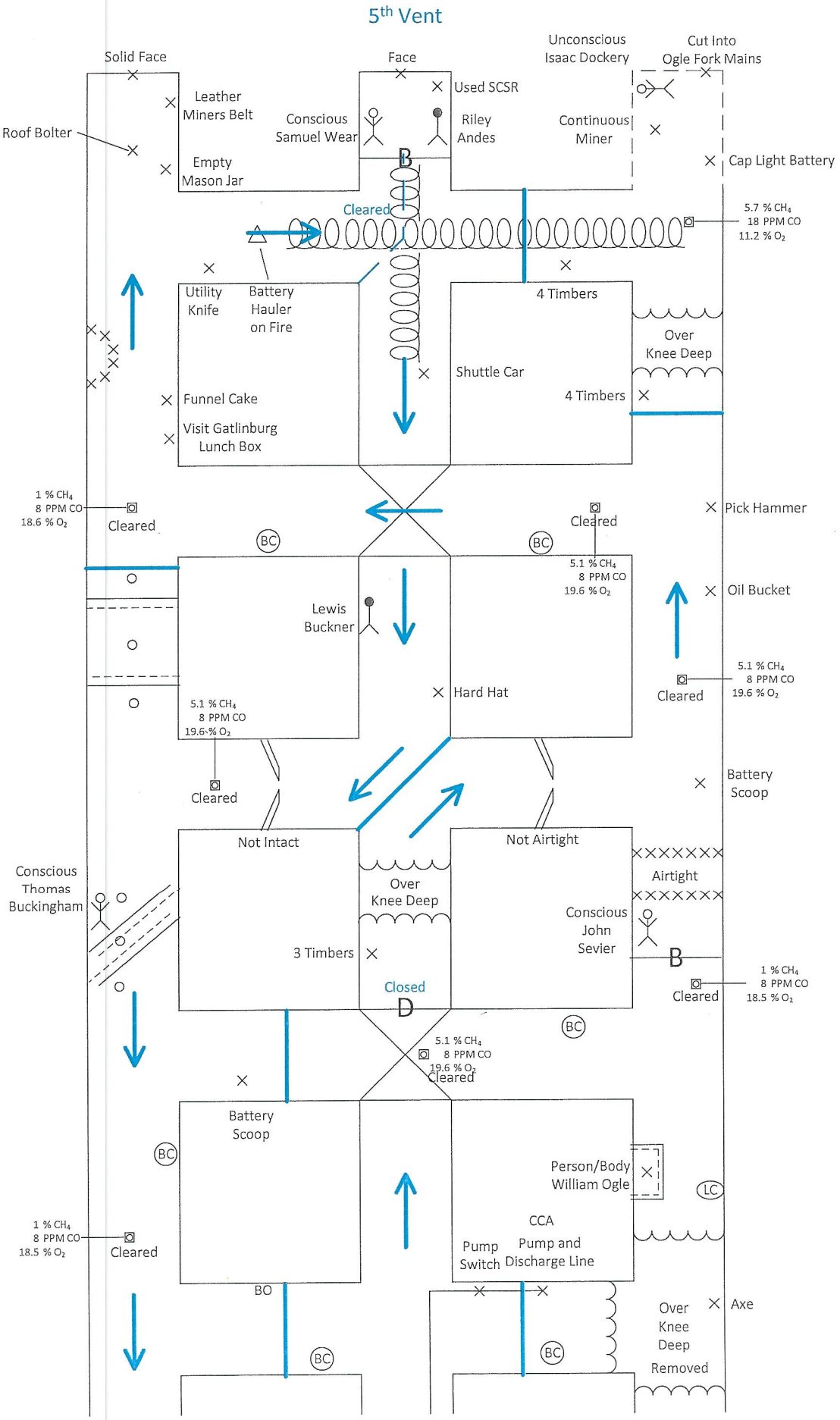
2nd Vent


Solid Face
Face
Unconscious
Isaac Dockery
Cut Into
Ogle Fork Mains
Leather
Miners Belt
Used SCSR
Roof Bolter
Conscious
Samuel Wear
Riley
Andes
Continuous
Miner
Empty
Mason Jar
B
Cap Light Battery
5.7 % CH4
18 PPM CO
11.2 % O2
Utility
Knife
Battery
Hauler
on Fire
4 Timbers
Over
Knee Deep
Shuttle Car
Funnel Cake
4 Timbers
Visit Gatlinburg
Lunch Box
1 % CH4
8 PPM CO
18.6 % O2
BC
BC
Pick Hammer
5.1 % CH4
8 PPM CO
19.6 % O2
Oil Bucket
Lewis
Buckner
5.1 % CH4
8 PPM CO
19.6 % O2
Hard Hat
5.1 % CH4
8 PPM CO
19.6 % O2
Battery
Scoop
Not Intact
Not Airtight
Conscious
Thomas
Buckingham
Over
Knee Deep
Airtight
Conscious
John
Sevier
3 Timbers
B
Closed
D
1 % CH4
8 PPM CO
18.5 % O2
Cleared
BC
5.1 % CH4
8 PPM CO
19.6 % O2
Cleared
Battery
Scoop
BC
Person/Body
William Ogle
LC
CCA
Pump
Switch
Pump and
Discharge Line
1 % CH4
8 PPM CO
18.5 % O2
Cleared
BO
Axe
Over
Knee
Deep
Removed
BC
BC

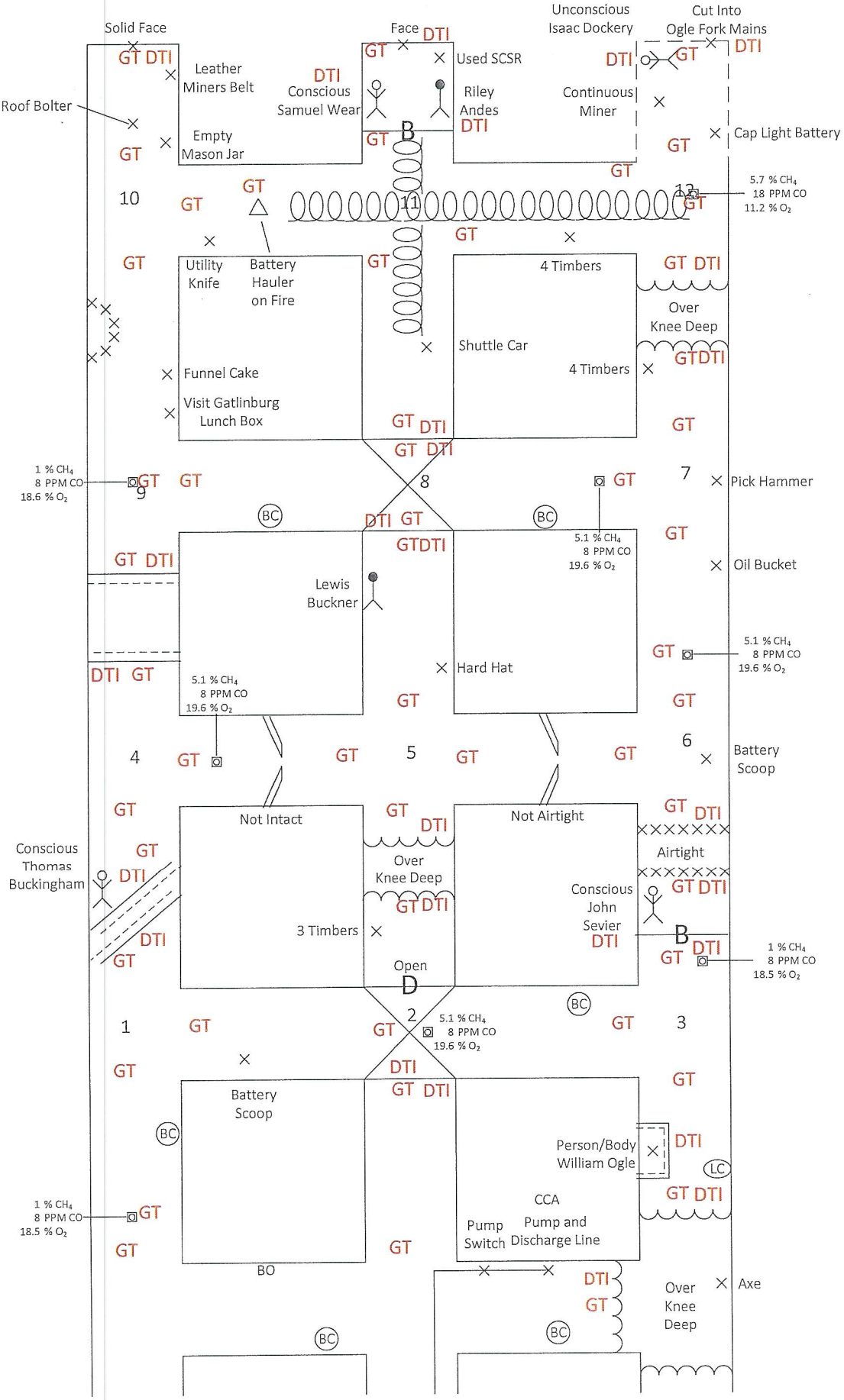

3rd Vent
Solid Face
Face
Unconscious
Isaac Dockery
Cut Into
Ogle Fork Mains
Leather
Miners Belt
Used SCSR
Roof Bolter
Conscious
Samuel Wear
Riley
Andes
Continuous
Miner
Cap Light Battery
Empty
Mason Jar
B
5.7 % CH4
18 PPM CO
11.2 % O2
Utility
Knife
Battery
Hauler
on Fire
4 Timbers
Over
Knee Deep
Shuttle Car
4 Timbers
Funnel Cake
Visit Gatlinburg
Lunch Box
1 % CH4
8 PPM CO
18.6 % O2
Pick Hammer
BC
BC
5.1 % CH4
8 PPM CO
19.6 % O2
Oil Bucket
Lewis
Buckner
5.1 % CH4
8 PPM CO
19.6 % O2
Hard Hat
5.1 % CH4
8 PPM CO
19.6 % O2
Cleared
Battery
Scoop
Not Intact
Not Airtight
Conscious
Thomas
Buckingham
Over
Knee Deep
Airtight
Conscious
John
Sevier
3 Timbers
B
Closed
D
1 % CH4
8 PPM CO
18.5 % O2
Cleared
BC
5.1 % CH4
8 PPM CO
19.6 % O2
Cleared
Battery
Scoop
BC
Person/Body
William Ogle
LC
CCA
Pump
Switch
Pump and
Discharge Line
1 % CH4
8 PPM CO
18.5 % O2
Cleared
BO
Axe
Over
Knee
Deep
Removed
BC
BC

4th Vent


Solid Face
Face
Unconscious
Isaac Dockery
Cut Into
Ogle Fork Mains
Leather
Miners Belt
Used SCSR
Roof Bolter
Conscious
Samuel Wear
Riley
Andes
Continuous
Miner
Empty
Mason Jar
B
Cap Light Battery
5.7 % CH4
18 PPM CO
11.2 % O2
Utility
Knife
Battery
Hauler
on Fire
4 Timbers
Over
Knee Deep
Shuttle Car
Funnel Cake
4 Timbers
Visit Gatlinburg
Lunch Box
1 % CH4
8 PPM CO
18.6 % O2
Cleared
Cleared
Pick Hammer
BC
BC
5.1 % CH4
8 PPM CO
19.6 % O2
Oil Bucket
Lewis
Buckner
Set Timbers
Hard Hat
5.1 % CH4
8 PPM CO
19.6 % O2
Cleared
5.1 % CH4
8 PPM CO
19.6 % O2
Cleared
Battery
Scoop
Not Intact
Not Airtight
Conscious
Thomas
Buckingham
Over
Knee Deep
Airtight
Timbers Set
3 Timbers
Conscious
John
Sevier
B
Closed
D
1 % CH4
8 PPM CO
18.5 % O2
Cleared
BC
5.1 % CH4
8 PPM CO
19.6 % O2
Cleared
Battery
Scoop
Person/Body
William Ogle
LC
BC
CCA
Pump
Switch
Pump and
Discharge Line
1 % CH4
8 PPM CO
18.5 % O2
Cleared
BO
Axe
Over
Knee
Deep
Removed
BC
BC


5th Vent
Solid Face
Face
Unconscious
Isaac Dockery
Cut Into
Ogle Fork Mains
Roof Bolter
Leather
Miners Belt
Conscious
Samuel Wear
Used SCSR
Riley
Andes
Continuous
Miner
Cap Light Battery
Empty
Mason Jar
B
Cleared
5.7 % CH4
18 PPM CO
11.2 % O2
Utility
Knife
Battery
Hauler
on Fire
4 Timbers
Over
Knee Deep
Shuttle Car
Funnel Cake
4 Timbers
Visit Gatlinburg
Lunch Box
1 % CH4
8 PPM CO
18.6 % O2
Cleared
Cleared
Pick Hammer
BC
BC
5.1 % CH4
8 PPM CO
19.6 % O2
Lewis
Buckner
Oil Bucket
Hard Hat
5.1 % CH4
8 PPM CO
19.6 % O2
Cleared
5.1 % CH4
8 PPM CO
19.6 % O2
Cleared
Battery
Scoop
Not Intact
Not Airtight
Conscious
Thomas
Buckingham
Over
Knee Deep
Airtight
Conscious
John
Sevier
3 Timbers
B
Closed
D
1 % CH4
8 PPM CO
18.5 % O2
Cleared
BC
5.1 % CH4
8 PPM CO
19.6 % O2
Cleared
Battery
Scoop
BC
Person/Body
William Ogle
LC
CCA
Pump
Switch
Pump and
Discharge Line
1 % CH4
8 PPM CO
18.5 % O2
Cleared
BO
Over
Knee
Deep
Removed
Axe
BC
BC

Problem


Solid Face
Face
Unconscious
Isaac Dockery
Cut Into
Ogle Fork Mains
Leather
Miners Belt
Used SCSR
Roof Bolter
Conscious
Samuel Wear
Riley
Andes
Continuous
Miner
Empty
Mason Jar
Cap Light Battery
5.7 % CH4
18 PPM CO
11.2 % O2
Utility
Knife
Battery
Hauler
on Fire
4 Timbers
Over
Knee Deep
Shuttle Car
Funnel Cake
4 Timbers
Visit Gatlinburg
Lunch Box
1 % CH4
8 PPM CO
18.6 % O2
Pick Hammer
Lewis
Buckner
5.1 % CH4
8 PPM CO
19.6 % O2
Oil Bucket
Hard Hat
Battery
Scoop
Not Intact
Not Airtight
Airtight
Conscious
Thomas
Buckingham
Over
Knee Deep
3 Timbers
Conscious
John
Sevier
Open
Battery
Scoop
Person/Body
William Ogle
CCA
Pump
Switch
Pump and
Discharge Line
BO
Over
Knee
Deep
Axe

Team Stops

Team Stop #1

- Teams will travel up the #1 entry to the first crosscut
- Captain will explore all openings DTI, R&R the unsafe
- Patient will give statement to teams "Help get me out there is bad roof in front of me". The team is now tied to the patient. But they do not have means to recover yet.

Team Stop #2

- Team will travel to the #2 entry
- Captain will make all openings DTI the overcast walls travel through the overcast door to water over knee deep and DTI.
- Team now has the means to recover the patient behind the unsafe. Team will travel back to the #1 entry and timber through the unsafe to recover the patient.

Team Stop #3

- Teams will travel to the #3 entry
- The captain will make all openings R&R the unsafe outby the intersection.
- Patient behind the barricade Gives Statement to team " Help its airtight behind me
- The team is now tied to the patient and have means to ventilate by pumping the water and exploring the intersection in the FAB
- See 1st and 2nd ventilation
- Teams cannot move the irrespirable over the person in the unsafe and they cannot move the Explosive mixture over the battery scoop
- Teams will recover the patient captain will DTI the Caved airtight.

Team Stop #4

- Teams will travel to the #1 entry second crosscut.
- The captain will make all openings. DTI, R&R the unsafe

Team Stop #5

- Teams will travel to the #2 entry second crosscut
- Captain will make all openings
- Captain will DTI body on left rib
- Captain will DTI overcast wall

Team Stop #6

- Teams will travel to #3 entry Second crosscut
- Captain will make all openings
- Captain will DTI the back side of the Caved Airtight

Team Stop #7

- Team will travel up the #3 entry to the 3rd crosscut
- Captain will make all openings
- Captain will DTI the water over knee deep

Team Stop #8

- Teams will travel from the #3 entry to the #2 entry 3rd crosscut
- Captain will DTI both overcast walls and make all openings

Team Stop #9

- Team will travel from the #2 entry to the #1 entry 3rd crosscut
- Captain will make all openings
- R&R the back side of the unsafe roof
- R&R the partial caved

Final Vent Map

